Name: Koushick S
Reg no: 1912260725

Experiment 1

Query and Output screenshot

MySQL 8.0 Command Line Cli

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
mysql> CREATE DATABASE IF NOT EXISTS dbms_lab;
Query OK, 1 row affected, 1 warning (0.04 sec)

mysql> USE dbms_lab;
Database changed
mysql>
mysql> CREATE TABLE Student (
    ->     id INT,
    ->     name VARCHAR(30)
    -> );
ERROR 1050 (42S01): Table 'student' already exists
mysql>
mysql> DESCRIBE Student;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| id         | int         | YES  |     | NULL    |       |
| name       | varchar(30) | YES  |     | NULL    |       |
| department | varchar(20) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql>
mysql> ALTER TABLE Student ADD department VARCHAR(20);
ERROR 1060 (42S21): Duplicate column name 'department'
mysql> DESCRIBE Student;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| id         | int         | YES  |     | NULL    |       |
| name       | varchar(30) | YES  |     | NULL    |       |
| department | varchar(20) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql>
mysql> INSERT INTO Student VALUES (1, 'Arun', 'AIML');
Query OK, 1 row affected (0.00 sec)

mysql>
```

Experiment 2

Query and Output screenshot

```
mysql> DROP TABLE IF EXISTS Student;
Query OK, 0 rows affected (0.06 sec)

mysql> DROP TABLE IF EXISTS Department;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE Department (
    ->     dept_id INT PRIMARY KEY,
    ->     dept_name VARCHAR(30) UNIQUE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE Student (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(30) NOT NULL,
    ->     age INT CHECK (age >= 17),
    ->     dept_id INT,
    ->     city VARCHAR(20) DEFAULT 'Chennai',
    ->     FOREIGN KEY (dept_id)
    ->         REFERENCES Department(dept_id)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> INSERT INTO Department VALUES (10, 'AIML');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO Student (id, name, age, dept_id)
    -> VALUES (1, 'Kavi', 18, 10);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+----+------+------+---------+---------+
| id | name | age  | dept_id | city    |
+----+------+------+---------+---------+
|  1 | Kavi |   18 |      10 | Chennai |
+----+------+------+---------+---------+
1 row in set (0.00 sec)

mysql>
```

Experiment 3

Query and Output screenshot

```
mysql>
mysql> DROP TABLE IF EXISTS Student;
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE Student (
    ->     id INT,
    ->     name VARCHAR(30),
    ->     course VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Student VALUES (1, 'Anu', 'AIML');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student VALUES (2, 'Bala', 'CSE');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+------+------+--------+
| id   | name | course |
+------+------+--------+
|    1 | Anu  | AIML   |
|    2 | Bala | CSE    |
+------+------+--------+
2 rows in set (0.00 sec)
```

Experiment 4

Query and Output screenshot

```
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO Student VALUES (1, 'Anu', 75);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student VALUES (2, 'Bala', 68);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> UPDATE Student
    -> SET marks = 85
    -> WHERE id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM Student;
+------+------+-------+
| id   | name | marks |
+------+------+-------+
|    1 | Anu  |    85 |
|    2 | Bala |    68 |
+------+------+-------+
2 rows in set (0.00 sec)

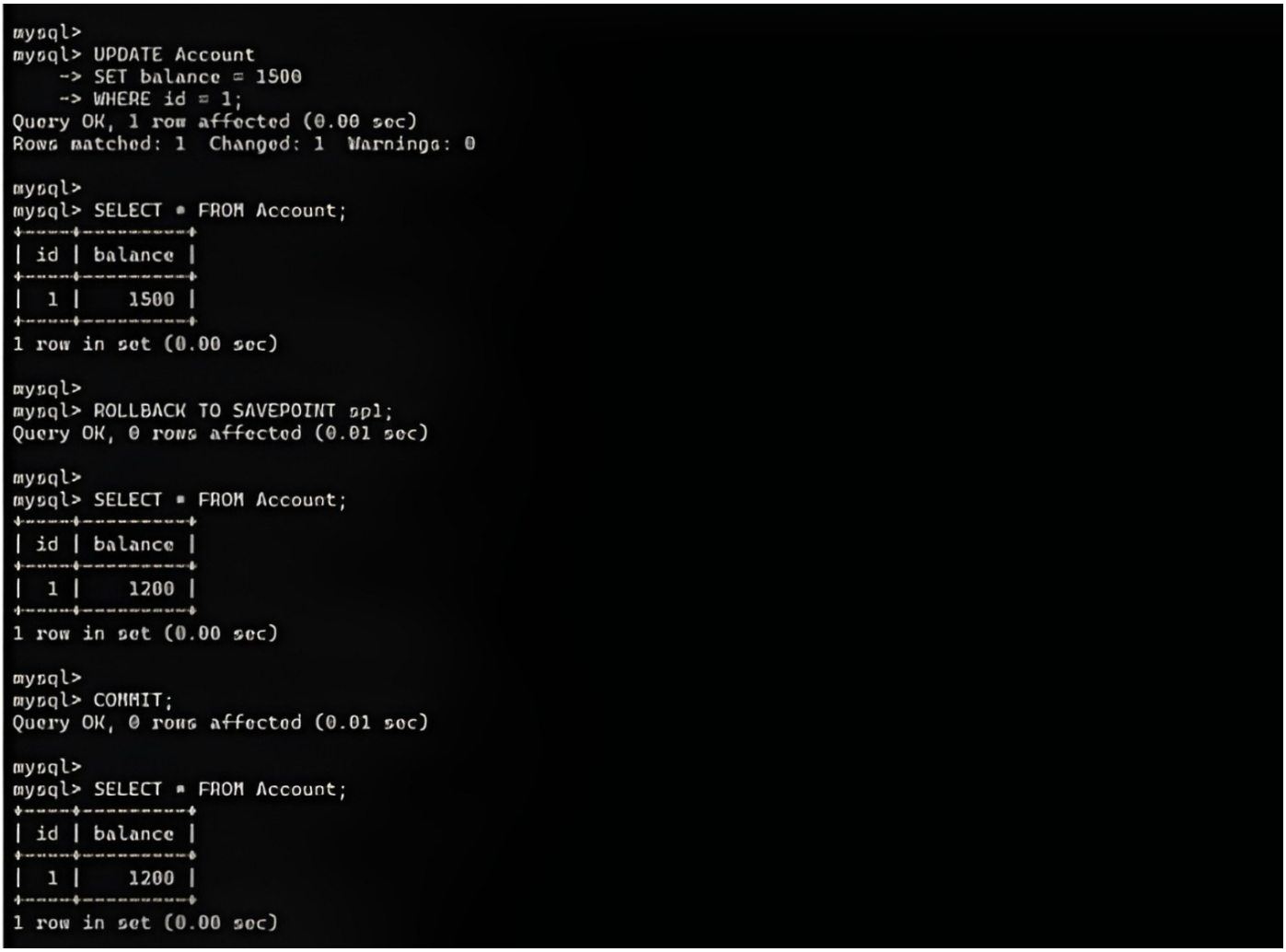
mysql>
mysql> DELETE FROM Student
    -> WHERE id = 2;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+------+------+-------+
| id   | name | marks |
+------+------+-------+
|    1 | Anu  |    85 |
+------+------+-------+
1 row in set (0.00 sec)

mysql>
```

Experiment 5

Query and Output screenshot



```
mysql>
mysql> UPDATE Account
    -> SET balance = 1500
    -> WHERE id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
|  1 |    1500 |
+----+---------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
|  1 |    1200 |
+----+---------+
1 row in set (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT * FROM Account;
+----+---------+
| id | balance |
+----+---------+
|  1 |    1200 |
+----+---------+
1 row in set (0.00 sec)
```